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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 20: EIA Case study-IV

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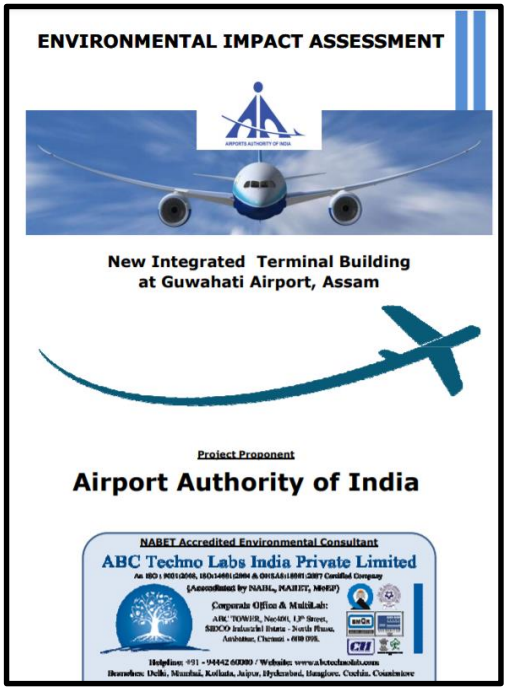


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EIA report of integrated terminal building at Guwahati airport, Assam

- New Integrated Terminal Building at Guwahati Airport, Assam
- Airport: Lokpriya Gopinath Bordoloi International Airport (LGBI) / Guwahati International Airport
- Project proponent: Airport Authority of India (AAI)
- EIA report is prepared by 'ABC Techno Labs India Private Limited'



Airport Authority of India (AAI)



- Airports Authority of India (AAI) was constituted by an Act of Parliament in 1995
- Entrusted with the responsibility of creating, upgrading, maintaining and managing civil aviation infrastructure in India
- AAI manages a total of 137 airports
- AAI provides air navigation services over 2.8 million square nautical miles of air space.

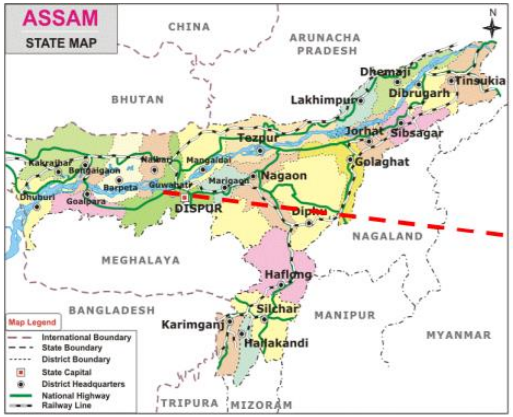
Guwahati Airport

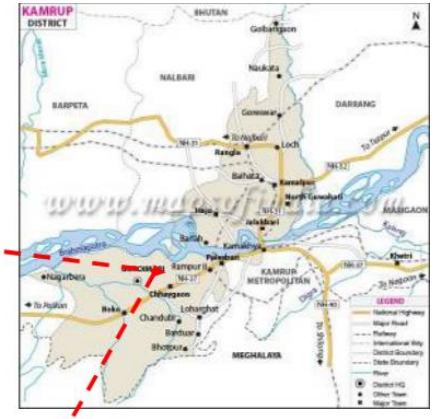
- One of the major airport in India and in North-East region of India, is situated in the state of Assam, belongs to AAI Operators
- Developed as inter-region hub
- Suitable for operation of medium size (B-767/A-310 type) aircrafts

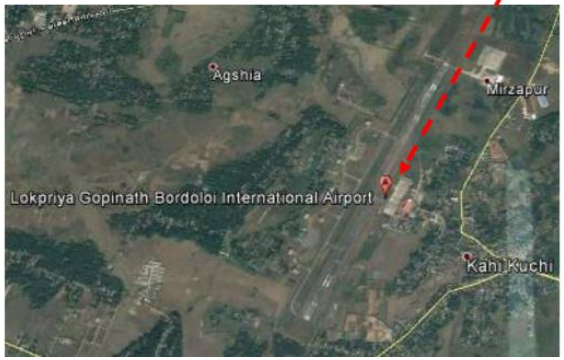


Image: North-East region of India

Index map of Guwahati airport







Guwahati Airport: Before project



- Terminal building at Guwahati Airport was handling 75 flights movements per day and 525 flights in a week
- Got saturated
- Land area covered 528.225 Acres

Image source: aera.gov.in

Consideration of alternative for project site

- The existing terminal building at Guwahati had saturated.
- In view of the future traffic growth at Guwahati Airport, there was an urgent requirement of construction of New Integrated Terminal Building
- The site for proposed new integrated terminal building was to be within airport.
- Therefore, no alternative site had been considered for new integrated terminal building.

Guwahati Airport: Planned project

- Centrally air-conditioned new integrated terminal building of modular design
- With all modern facilities and associated facilities like multilevel car parking
- Designed for 2900 domestic and 200 international passengers at a time
- Level of service “B” as per IATA (International Air Transport Association) recommendations in initial years & level of service ”C” in year of saturation



Image source: worldarchitecture.org

Level of service for airports

Level	Standard
A	Excellent comfort
B	High comfort
C	Good comfort
D	Adequate comfort
E	Inadequate comfort
F	Unacceptable

Level of service depends on:

- Space per passenger
- Maximum waiting time

Time to time change in these standards by IATA (International Air Transport Association)

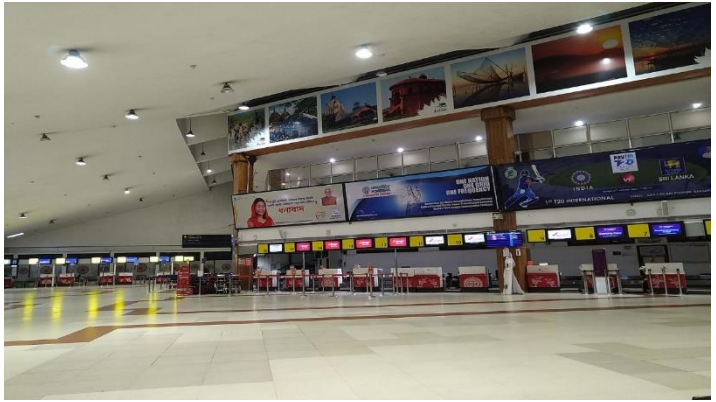


Image: Terminal building of Guwahati airport

Source: airport-technology.com

Benefits of the project

- Better infrastructure facilities for passengers
- Promotion of tourism, trade, commerce
- Generation of more revenue to the state
- More employment opportunity to people
- More business and industrial opportunities

Requirement of EIA for Environment Clearance (EC)

As per EIA notification 2006;

- All projects including airstrip, which are for commercial use require environment clearance from Ministry of Environment, Forest & Climate Change (MoEF&CC).
- Considered as category ‘A’ project (with high environmental impacts).
- Air strips which do not involve bunkering/ refueling facility and or Air Traffic Control, are exempted.

Other environmental regulations

Indian aviation sector follows norms of International Civil Aviation Organisation (ICAO):

- Reducing noise (noise stringency limits engines)
- Land use planning and management
- Reducing emissions at source (emission limits)
- Operational mechanism for reducing aviation emissions
- Market based options (legal & administrative issues surrounding emissions permit trading, environmental charges and voluntary agreements to limit or reduce emissions)

Community Consultation

- Group discussions was done among local population.
- Issues related to employment generation, information flow, health and safety, social welfare activity etc. were discussed.
- People expect preference should be given to local people for employment opportunities in various activities during construction phase





Impact Assessment Rating Matrix

Negligible Impact : Magnitude of change comparable to natural variation

Minor Impact : Detectable but not significant

Moderate Impact : Insignificant; amenable to mitigation; should be mitigated where practicable

Major Impact : Defined as significant; amenable to mitigation; must be mitigated

Topography & physiography: Impact level & mitigation measures

- Construction site would be designed to minimize filling of the earths
- Borrowing of earth will be ensured only from approved borrow area
- Borrow area will be rehabilitated after borrowing of necessary earth.

Environmental Impact Rating	Topography & Physiography
Nature of Impact	Adverse
Duration of Impact	Long term
Impacted Area	Localized
Likelihood of Occurrence	Medium
Severity of Impact	Medium
Significance of Impact	Low

Land use: Impact level & mitigation measures

- Land clearing for construction site will be kept to the minimum practicable
- The filling and cutting of soil would be kept minimum
- Construction debris and waste generated during construction activities will be collected and disposed in environmentally sound manner

Environmental Impact Rating	Land Use pattern
Nature of Impact	Adverse
Duration of Impact	Long term
Impacted Area	Localized
Likelihood of Occurrence	Low
Severity of Impact	Slight
Significance of Impact	Negligible

Water resource: Impact level & mitigation measures

Total water requirement is estimated as 1010 kld (Fresh Water 710 kld)

Environment Impact Rating	Water Resources
Nature of Impact	Adverse
Duration of Impact	Long term
Impacted Area	Localized
Likelihood of Occurrence	High
Severity of Impact	Medium
Significance of Impact	Medium

- Efforts will be made to reduce water consumption
- Sewage and domestic wastewater will be treated in sewage treatment plant
- Treated wastewater will be reused in HVAC (Heating, ventilation & Air Conditioning), flushing, greenery and landscaping

Kld: Kilo Liters per day

Water quality: Impact level & mitigation measures

Total estimated wastewater generation during the operation phase is 808 kld (kilo liter per day)

Environmental Impact Rating	Water Quality
Nature of Impact	Adverse
Duration of Impact	Long term
Impacted Area	Localized
Likelihood of Occurrence	High
Severity of Impact	Medium
Significance of Impact	Medium

- Collection of wastewater and treatment in Sewage Treatment Plant (STP)
- Efficient operation of STP will be ensured
- Avoid spillage of fuel and lube oil and storing them on concrete floor
- Solid waste collection and proper disposal
- Regular testing and analysis of treated wastewater and compliance of discharge standards.

Air quality: Emissions and level of impact

Sources	Emission Rate of Pollutants (g/s)				
	SO_2	NOx	CO	HC	PM
Aircraft	5.62	4.80	47.39	70.30	--
DG set	2.92	3.26	--	---	0.54
Vehicular	--	1.87	3.27	0.85	0.32

Environmental Impact Rating	Air Quality
Nature of Impact	Adverse
Duration of Impact	Long term
Impacted Area	Localized
Likelihood of Occurrence	High
Severity of Impact	Medium
Significance of Impact	Moderate

Air quality: Mitigation measures

- Compliance of all standards prescribed by the International Civil Aviation Organization (ICAO) during operation of aircrafts
- 30 m Stack height of DG sets
- No traffic congestion in front of new terminal building, it will help in reduction of vehicular emissions from the airport
- Ground vehicles at the airport will be maintained and have a "Pollution Under Control" certificate
- Development of greenery and landscaping at the airport for improving ambient air quality

Noise: Impact area & level of impact

Sr. No.	Noise Levels	Area of Influence (km^2)	Remark
1.	80.0	0.07	Predicted noise levels confined within the airport boundary
2.	75.0	0.21	Predicted noise levels confined within the airport boundary
3.	65.0	1.94	The predicted noise levels fall within the airport boundary
4.	55.0	5.32	About 2.4 km^2 of area with these noise level is out side the airport
5.	50.0	8.63	About 6.7 km^2 of area with these noise level is out side the airport

Impact Rating	Noise Level
Nature of Impact	Adverse
Duration of Impact	Long term
Impacted Area	Localized
Likelihood of Occurrence	High
Severity of Impact	Low
Significance of Impact	Low

Noise pollution: Mitigation measures

- No traffic congestion at the airport to reduce vehicular noise emissions from the airport
- DG sets have been provided with acoustic enclosure
- Terminal building will be soundproof
- Ground staff wears earplug while attending the aircraft
- Green belt, landscaping and boundary at the airport act as barrier for noise
- Green belt/plantation in the nearby settlements

Terrestrial ecology: Impacts level & mitigation measures

For proposed new terminal building no trees felling will be required

Environmental Impact Rating	Terrestrial Ecology
Nature of Impact	Beneficial
Duration of Impact	Long term
Impacted Area	Localized
Likelihood of Occurrence	High
Severity of Impact	Low
Significance of Impact	Minor

Mitigation measures

- Landscaping/ plantation/greenery will be increased after new terminal building

- Indigenous species of trees will be planted after new terminal building

Impacts on Deepor Beel bird sanctuary

- Deepor Beel Bird Sanctuary, a Ramsar site is located at 3 km from the project site.
- It is not located in funnel area of runway of Guwahati Airport.
- Noise modelling carried for existing airport, indicate that impact of aircraft noise from aircraft operation on the Deepor Beel Bird Sanctuary is negligible.

Funnel area:

- Imaginary flight path
- Extending outwards and upwards from either side of runway
- Generally, up to 15 degree of runway

Precautionary mitigation measures for Deepor Beel bird sanctuary

- Landscaping and green belt will be developed towards Deepor Beel bird sanctuary.

- Recommendations of Chief Wildlife Warden will be implemented (if any in future)



Image: Deepor Beel bird sanctuary

Image source: telgraphindia

Employment & growth: Impacts & impact level

- Boost in tourism and commercial activities in the region.
- Improve direct and indirect employment opportunities
- Revenue generation, commercial and industrial activities

Environmental Impact Rating	Employment & Economic Growth
Nature of Impact	Beneficial
Duration of Impact	Long term
Impacted Area	Regional
Likelihood of Occurrence	High
Severity of Impact	Medium
Significance of Impact	Moderate

Energy conservation measures

GRIHA: Green Rating for Integrated Habitat Assessment
BEE: Bureau of Energy Efficiency (BEE) India

- Terminal building designed and constructed for GRIHA Rating 4 star
- Energy efficient building material & glass
- LED lamps instead of GLS (General Lighting Services) lamps
- Energy efficient HVAC (Heating, Ventilation & Air Conditioning) system
- Use of 5-star BEE energy efficiency rating electrical equipment

Energy conservation measures

- Microprocessor-based Building Management System (BMS) for minimization of energy consumption
- Automatic lighting on/ off control system will be provided in the airport area for optimum utilization of energy.
- 540 KW solar power generation plant will be established at the airport to produce clean energy.

By adopting above measures about 30% energy will be saved

Monitoring

Environment component	Project stage	Parameters	Location	Frequency
Air	Construction	PM2.5, PM10, SO2, NO2, CO	project site	24 hourly. once in season
	Operation	PM2.5, PM10, SO2, NO2, CO	project site	24 hourly once in season
Water quality	Construction	as per IS 10500:2012	project site	Half-yearly
	Operation	as per IS 10500:2013	project site	Half-yearly
Noise	Construction	Day & nighttime noise in dB(A)	4 location on airport boundary	Once in season
	Operation	Day & nighttime noise in dB(A)	Arrival & departure location, boundary of airport	Once in season
Treated waster water from STP	Operation	pH, TDS, TSS, BOD, COD, Oil & grease	Outlet of STP	Once in month

Environmental reporting

- Compliance monitoring and the progress reports monthly during the implementation period.
- The operation stage monitoring reports may be half yearly or annually
- Monitoring will primarily ensure that actions taken are in accordance with contract
- The photographic records will also be established to provide useful environmental monitoring tools

Potential risks for project

Potential hazards

- Storage of HSD for DG sets operation in underground tanks (leakage)
- Fire in ATF Tanks
- Crashing of aircrafts while landing or take-off
- Leakage of flammable materials, like, HSD followed by fire
- Natural calamities like, earthquake, cyclone, high winds, etc.

Potential external threats

- Agitation/forced entry by external group of people
- Sabotage
- Air raids
- Bomb threat at terminal building, cargo terminal and aircraft

HSD: High speed diesel
ATF: Aviation turbine fuel

Risk mitigations for accidental release of flammable liquid

Prompt action in the event of an accidental release of HSD or ATF is essential.

HSD: High speed diesel
ATF: Aviation turbine fuel

Provisions for a flammable liquid spill:

- Non-flammable absorbent material is available for immediate use
- Ignition sources can be quickly removed
- The area is well ventilated
- Spread of the spill is limited

Conclusion

- All possible impacts of projects are assessed, and suitable mitigation measures are provided.
- Periodic environmental impact monitoring process is put in place
- All the possible risks included in EIA report and necessary provisions are suggested

References:

- Airport Authority of India (AAI), 2016 “Environmental Impact Assessment report for New Integrated Terminal Building at Guwahati Airport, Assam”

Thank You